

## Review Form – NLP AI 2026

Paper ID: XA2031

Paper Title: SchemaRAG-SQL: A Multi-Agent Retrieval-Augmented Framework for Reliable and Privacy-Preserving Text-to-SQL Generation

### ● Evaluation Summary

| Evaluation Criteria                      | Score (1 = Low / Poor, 5 = High / Excellent) |
|------------------------------------------|----------------------------------------------|
| 1. Relevance to Conference Themes        | 4                                            |
| 2. Clarity and Organization of the Paper | 4                                            |
| 3. Scientific Merit / Originality        | 3                                            |
| 4. Technical Quality                     | 3                                            |
| 5. Figures and Tables                    | 3                                            |
| 6. Language and Style                    | 3                                            |
| 7. References and Background             | 3                                            |

### ● Overall Recommendation:

- ☐ Accept
- ☐ Accept with Minor Revision
- ☒ **Accept with Major Revision**
- ☐ Reject

### ● Comments and Suggestions for Improvement:

1. The paper addresses a genuinely important deployment problem — privacy-preserving, fully local Text-to-SQL — and the system design is coherent and well-motivated. However, the experimental evaluation has critical methodological gaps, the primary metric (EAc) is not validated, and several claims in the abstract and introduction are overstated or contradicted by the results.
2. The paper's central accuracy claim rests on EAc — a proxy for execution accuracy defined by the authors — which has never been validated against true execution accuracy on any subset. Reporting an unvalidated proxy as the headline metric in a research paper, without any cross-validation, is insufficient for publication. At minimum, the authors must run true execution accuracy on a randomly sampled subset (e.g., 100 queries) and report how closely EAc tracks it.
3. Table 1 reports the Full pipeline achieving SVR = 100% and EAc = 13.10%, while Table 2 shows the No-Validator setting reaching SVR = 99.8% and EAc = 13.67%. The No-Validator configuration should not outperform the Full pipeline. Additionally, RAG-only SVR (97%) conflicts with the

higher SVR (99.8%) in Table 2. These inconsistencies are not explained and weaken the ablation analysis. The abstract's "+3 pp" claim for RAG also contradicts Table 1 (+15 pp).

4. Table 2 shows that adding the Validator decreases EAc (13.67% → 12.80%) and EMn (13.17% → 12.53%). This indicates that the validator may harm overall accuracy despite reducing syntax errors. This limitation is under-analyzed and contradicts the paper's positive framing.
5. The paper does not compare against other local or open-weight models.
6. The NV evaluation uses 600 samples across 12 databases; the WV evaluation uses 375 samples across 6 databases — different databases, different samples, different sizes. These are not matched conditions. Comparing NV and WV performance across these splits is not a valid within-system comparison. Both configurations should be evaluated on the same set of queries.
7. Section 3.3 contains an incomplete sentence: "At most three attempts are made; if all" — the sentence ends abruptly. This appears to be a copyediting error and must be completed.
8. The acknowledgements section contains grammatical errors: "in her exemplary. leadership" contains a spurious period. "and Their open-source Qwen2.5" has an erroneous capital. These suggest the paper was not carefully proofread before submission.
9. The paper cites "[23]" for the all-MiniLM-L6-v2 embedder, but reference [23] is the ChromaDB citation (Chroma Inc.).
10. The claim of being the "first paper to formally define and measure SVR" is not verified. This is stated in the Contributions section. Given that syntactic validity is a commonly discussed property in the Text-to-SQL literature, the authors should either provide a literature survey confirming this claim or soften the language to "to the best of our knowledge."